AAE 451 Aircraft Conceptual Design

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CONTENTS

Welcome to the AAE 451 Aircraft Conceptual Design Jupyter Book!

> [!TIP] ? **Download Complete Textbook PDF (LaTeX Version)**

This book consolidates **Assignments 1 through 12** into a single, continuous interactive format. It covers the full scope of the aircraft conceptual design process, including:

- **Assignment 1 & 2:** Baseline Data & Mission Requirements
- **Assignment 3:** Initial Gross Weight Sizing
- **Assignment 4:** Aerodynamic Analysis
- **Assignment 5 & 6:** Propulsion & Constraint Analysis
- **Assignment 7 & 8:** Configuration & Tail Sizing
- **Assignment 9:** Performance Assessment
- **Assignment 10 & 11:** Structural Analysis & Wing Box Design
- **Assignment 12:** Dynamic Stability

Use the table of contents on the left to navigate through each stage of the design process.

CHAPTER

ONE

ASSIGNMENT 1 - BASELINE CHARACTERIZATION & GAP ANALYSIS

Reference: 2025-2026 Undergraduate Team Aircraft Design RFP

Topic: Baseline Study and Historical Data Gathering

1.1 1. Overview

The Request for Proposal (RFP) identifies the **F/A-18E/F Super Hornet** as the baseline aircraft your design must replace. To build a better aircraft, you must first understand the limitations and capabilities of the incumbent.

This assignment establishes your design benchmark. You will gather historical and technical data to create a "Baseline Comparison Matrix" and identify the specific performance gaps your new design must close.

1.2 2. Objectives

- **Quantify** key geometric and performance parameters of the F/A-18E/F Super Hornet.
- **Verify** the accuracy of the RFP data (specifically the carrier suitability tables).
- **Identify** performance gaps between current fleet capabilities and the new requirements.
- **Define** the geometric constraints (spot factor, deck handling) that will drive your initial sizing.

1.3 3. Required Tasks

1.3.1 Task A: Data Verification Audit

The RFP provides a "Catalog of Current Carrier Suitability Data" in **Section 5, Table 3.1.10-II** (Page 8). This table lists an entry for "F/A-18".

1. Locate independent, reliable sources for the **F/A-18E/F Super Hornet** (e.g., Navy Fact Files, SAR reports).
2. Compare the **Length** and **Gross Weight** in the RFP table against your independent sources.
3. Determine if the RFP table describes the *Super Hornet* (E/F) or the *Legacy Hornet* (A/C).
4. If the RFP data is outdated, derive the correct values for the Super Hornet. **Do not use incorrect baseline data for your design.**

1.3.2 Task B: Detailed Geometric Characterization

Collect the detailed sizing data required to generate an initial CAD model. Populate **Table 1** with the following:

- **Main Wing:** Reference Area (S_{ref}), Aspect Ratio (AR), Taper Ratio (λ), Sweep angles (Λ_{LE}, $\Lambda_{c/4}$), Airfoil types, Incidence, and Dihedral.
- **Empennage:** Areas, spans, sweeps, taper ratios, and tail arms (L_{ht}, L_{vt}) for both horizontal and vertical stabilizers.
- **Fuselage:** Total length (compare to the 50 ft RFP limit), max width/height, and cross-section shape.
- **Landing Gear:** Wheelbase, track width, tire sizes, strut lengths, and tip-back angle.

1.3.3 Task C: Performance Gap Analysis

Populate **Table 2** to compare the Super Hornet against the RFP requirements. Identify the most difficult constraints.

- **Range:** Compare the 700 nm requirement against the Super Hornet's actual internal-fuel combat radius.
- **Speed:** Compare the Mach 2.0 dash speed goal against the Super Hornet's top speed.
- **Approach:** Compare the <145 kt approach speed requirement against the baseline.

1.3.4 Task D: Comparable Aircraft Survey

Research the aircraft listed in **Table 3**. These platforms represent successful design solutions for specific requirements relevant to this competition (e.g., high efficiency, variable geometry, stealth). Populate the table with their high-level parameters.

1.3.5 Task E: Comparative Wing Analysis

Using the aircraft from Task D, analyze how mission roles influence wing design. Populate **Table 4**.

- Compare **Aspect Ratio (AR)** versus **Sweep Angle** across different eras.
- Calculate **Wing Loading** (W_{MTOW}/S) for each aircraft. This metric is critical for estimating approach speeds.

1.4 4. Deliverables

1. Slides Submit a maximum of 4 professional slides containing:

- **Executive Summary:** State the most significant design challenge found (e.g., length constraints vs. baseline size).
- **Data Audit Result:** Definitive statement on the RFP table accuracy and the corrected values used.
- **Gap Analysis:** Brief discussion of the primary performance driver (Range, Speed, or Payload).
- **Attachments:** Your completed Tables 1, 2, 3, and 4.

2. Data Tables (.csv or .xlsx)

- Submit the completed spreadsheet files.

1.5 5. Resources

- **RFP Link 2:** *F/A-18E/F Super Hornet Aircraft (F/A-18E/F) SAR*
- **RFP Link 3:** *Lessons Learned from the F/A-18E/F Development Programs*
- **Official Navy Fact File:** Navy.mil F/A-18E/F Page

1.6 Appendix: Data Templates

Run the code block below to generate the blank data tables. You can fill them in programmatically or export them to Excel/CSV.

```python
import pandas as pd
import numpy as np
from IPython.display import display, Markdown

# ==========================================
# TABLE GENERATION
# ==========================================

def get_detailed_geometry_template():
    """Generates Table 1: Detailed Geometric Parameters"""
    data = [
        ("Overall Length", "ft", "<= 50.0", "", ""),
        ("Wingspan (Unfolded)", "ft", "<= 60.0", "", ""),
        ("Wingspan (Folded)", "ft", "<= 35.0", "", ""),
        ("Max Takeoff Weight", "lb", "<= 90,000", "", ""),
        ("Fineness Ratio (L/D)", "-", "N/A", "", ""),
        ("Max Cross-Section Width", "ft", "N/A", "", ""),
        ("Max Cross-Section Height", "ft", "N/A", "", ""),
        ("Wing Reference Area (S)", "ft^2", "N/A", "", ""),
        ("Wing Aspect Ratio (AR)", "-", "N/A", "", ""),
        ("Wing Sweep (Leading Edge)", "deg", "N/A", "", ""),
        ("Wing Sweep (Quarter Chord)", "deg", "N/A", "", ""),
        ("Wing Taper Ratio", "-", "N/A", "", ""),
        ("Wing Airfoil (Root / Tip)", "Type", "N/A", "", ""),
        ("Wing Dihedral", "deg", "N/A", "", ""),
        ("Horiz. Tail Area", "ft^2", "N/A", "", ""),
        ("Horiz. Tail Sweep (LE)", "deg", "N/A", "", ""),
        ("Horiz. Tail Taper Ratio", "-", "N/A", "", ""),
        ("Horiz. Tail Arm (Wing c/4 to HT c/4)", "ft", "N/A", "", ""),
        ("Vert. Tail Area (per side)", "ft^2", "N/A", "", ""),
        ("Vert. Tail Height", "ft", "N/A", "", ""),
        ("Vert. Tail Cant Angle", "deg", "N/A", "", ""),
        ("Vert. Tail Taper Ratio", "-", "N/A", "", ""),
        ("Vert. Tail Arm (Wing c/4 to VT c/4)", "ft", "N/A", "", ""),
        ("Landing Gear Type", "-", "Tricycle", "", ""),
        ("Landing Gear Wheelbase", "ft", "N/A", "", ""),
        ("Landing Gear Track Width", "ft", "N/A", "", ""),
        ("Main Tire Size (Dia x Width)", "in", "N/A", "", ""),
        ("Nose Tire Size (Dia x Width)", "in", "N/A", "", ""),
        ("LG Tip-Back Angle", "deg", "> 15 (typ)", "", ""),
    ]
```

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```
    return pd.DataFrame(data, columns=["Parameter", "Unit", "RFP Limit/Ref", "F/A-18E/
↪F Baseline (Actual)", "Source"])

def get_performance_template():
    """Generates Table 2: Performance Gap Analysis"""
    data = [
        ("Wind Over Deck (WOD)", "Tropical Day", "0 kts", "", ""),
        ("Launch Endspeed", "Max Lift", "See C-13-2 Curves", "", ""),
        ("Max Speed (Vmax)", "30k ft", "Mach 1.6 (Req)", "", ""),
        ("Combat Radius (Strike)", "Internal Fuel", ">= 700 nm", "", ""),
        ("Sustained Turn Rate", "20k ft", ">= 8.0 deg/sec", "", ""),
        ("Vertical Load Factor", "Mid Mission", ">= 7.0 g", "", ""),
        ("Approach Speed", "Landing Wt", "< 145 kts", "", ""),
        ("Arresting WOD", "Max", "<= 15 kts", "", ""),
        ("Single Engine ROC", "Approach", ">= 500 ft/min", "", ""),
    ]
    return pd.DataFrame(data, columns=["Metric", "Condition", "RFP Requirement", "F/A-
↪18E/F Baseline (Actual)", "Source"])

def get_comparable_aircraft_table_3():
    """Generates Table 3: Comparable Aircraft Survey"""
    data = [
        ("F-35C Lightning II", "Strike Fighter (Stealth)", "2019", "", "", "", ""),
        ("Dassault Rafale M", "Carrier Strike", "2001", "", "", "", ""),
        ("Grumman A-6 Intruder", "Long Range Strike", "1963", "", "", "", ""),
        ("LTV A-7 Corsair II", "Light Attack", "1967", "", "", "", ""),
        ("Grumman F-14 Tomcat", "Fleet Defense", "1974", "", "", "", ""),
        ("Sukhoi Su-33", "Carrier Fighter", "1998", "", "", "", ""),
    ]
    return pd.DataFrame(data, columns=["Aircraft", "Role", "IOC Year", "MTOW (lb)",
↪"Max Speed", "Radius (nm)", "Source"])

def get_wing_comparison_table_4():
    """Generates Table 4: Comparative Wing Analysis"""
    columns = ["Parameter", "F/A-18E/F (Baseline)", "F-35C", "Rafale M", "A-6 Intruder
↪", "A-7 Corsair II", "F-14 Tomcat", "Su-33"]
    data = [
        ("Reference Area (S) [ft^2]", "", "", "", "", "", "", ""),
        ("Aspect Ratio (AR) [-]", "", "", "", "", "", "", ""),
        ("Wing Span (b) [ft]", "", "", "", "", "", "", ""),
        ("Taper Ratio (λ) [-]", "", "", "", "", "", "", ""),
        ("Sweep (Leading Edge) [deg]", "", "", "", "", "", "", ""),
        ("Thickness-to-Chord (t/c) [%]", "", "", "", "", "", "", ""),
        ("Wing Loading (W/S) [lb/ft^2]", "", "", "", "", "", "", ""),
    ]
    return pd.DataFrame(data, columns=columns)

# ==========================================
# DISPLAY & EXPORT
# ==========================================

# Generate empty tables
df_geo = get_detailed_geometry_template()
df_perf = get_performance_template()
df_comp = get_comparable_aircraft_table_3()
df_wing = get_wing_comparison_table_4()
```

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```
# Set Index to remove the generic index number (0, 1, 2...)
df_geo.set_index('Parameter', inplace=True)
df_perf.set_index('Metric', inplace=True)
df_comp.set_index('Aircraft', inplace=True)
df_wing.set_index('Parameter', inplace=True)

# Remove the index name to eliminate the extra header row
df_geo.index.name = None
df_perf.index.name = None
df_comp.index.name = None
df_wing.index.name = None

# Display formatted tables
display(Markdown("### Table 1: Detailed Geometric Characterization"))
display(df_geo)
print("\n" + "="*80 + "\n")

display(Markdown("### Table 2: Performance Gap Analysis"))
display(df_perf)
print("\n" + "="*80 + "\n")

display(Markdown("### Table 3: Comparable Aircraft Survey"))
display(df_comp)
print("\n" + "="*80 + "\n")

display(Markdown("### Table 4: Comparative Wing Analysis"))
display(df_wing)

# Optional: Uncomment to export
# df_geo.to_csv('01_Geometry_Detailed.csv')
# df_perf.to_csv('02_Performance_Gap.csv')
# df_comp.to_csv('03_Comparable_Aircraft.csv')
# df_wing.to_csv('04_Wing_Comparison.csv')
```

Table 1: Detailed Geometric Characterization

```
                                     Unit RFP Limit/Ref  \
Overall Length                         ft        <= 50.0
Wingspan (Unfolded)                    ft        <= 60.0
Wingspan (Folded)                      ft        <= 35.0
Max Takeoff Weight                     lb      <= 90,000
Fineness Ratio (L/D)                    -           N/A
Max Cross-Section Width                ft           N/A
Max Cross-Section Height               ft           N/A
Wing Reference Area (S)              ft^2           N/A
Wing Aspect Ratio (AR)                  -           N/A
Wing Sweep (Leading Edge)             deg           N/A
Wing Sweep (Quarter Chord)            deg           N/A
Wing Taper Ratio                        -           N/A
Wing Airfoil (Root / Tip)            Type           N/A
Wing Dihedral                         deg           N/A
Horiz. Tail Area                     ft^2           N/A
Horiz. Tail Sweep (LE)                deg           N/A
Horiz. Tail Taper Ratio                 -           N/A
Horiz. Tail Arm (Wing c/4 to HT c/4)   ft           N/A
```

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```
Vert. Tail Area (per side)            ft^2          N/A
Vert. Tail Height                       ft          N/A
Vert. Tail Cant Angle                  deg          N/A
Vert. Tail Taper Ratio                   -          N/A
Vert. Tail Arm (Wing c/4 to VT c/4)     ft          N/A
Landing Gear Type                        -     Tricycle
Landing Gear Wheelbase                  ft          N/A
Landing Gear Track Width                ft          N/A
Main Tire Size (Dia x Width)            in          N/A
Nose Tire Size (Dia x Width)            in          N/A
LG Tip-Back Angle                      deg    > 15 (typ)

                                     F/A-18E/F Baseline (Actual) Source
Overall Length
Wingspan (Unfolded)
Wingspan (Folded)
Max Takeoff Weight
Fineness Ratio (L/D)
Max Cross-Section Width
Max Cross-Section Height
Wing Reference Area (S)
Wing Aspect Ratio (AR)
Wing Sweep (Leading Edge)
Wing Sweep (Quarter Chord)
Wing Taper Ratio
Wing Airfoil (Root / Tip)
Wing Dihedral
Horiz. Tail Area
Horiz. Tail Sweep (LE)
Horiz. Tail Taper Ratio
Horiz. Tail Arm (Wing c/4 to HT c/4)
Vert. Tail Area (per side)
Vert. Tail Height
Vert. Tail Cant Angle
Vert. Tail Taper Ratio
Vert. Tail Arm (Wing c/4 to VT c/4)
Landing Gear Type
Landing Gear Wheelbase
Landing Gear Track Width
Main Tire Size (Dia x Width)
Nose Tire Size (Dia x Width)
LG Tip-Back Angle
```

```
================================================================================
```

Table 2: Performance Gap Analysis

```
                            Condition    RFP Requirement  \
Wind Over Deck (WOD)      Tropical Day              0 kts
Launch Endspeed               Max Lift  See C-13-2 Curves
Max Speed (Vmax)                30k ft     Mach 1.6 (Req)
Combat Radius (Strike)   Internal Fuel          >= 700 nm
Sustained Turn Rate             20k ft     >= 8.0 deg/sec
Vertical Load Factor       Mid Mission           >= 7.0 g
```

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```
Approach Speed              Landing Wt          < 145 kts
Arresting WOD                      Max         <= 15 kts
Single Engine ROC             Approach      >= 500 ft/min

                       F/A-18E/F Baseline (Actual) Source
Wind Over Deck (WOD)
Launch Endspeed
Max Speed (Vmax)
Combat Radius (Strike)
Sustained Turn Rate
Vertical Load Factor
Approach Speed
Arresting WOD
Single Engine ROC
```

```
================================================================================
```

Table 3: Comparable Aircraft Survey

```
                                          Role IOC Year MTOW (lb) Max Speed  \
F-35C Lightning II    Strike Fighter (Stealth)     2019
Dassault Rafale M               Carrier Strike     2001
Grumman A-6 Intruder         Long Range Strike     1963
LTV A-7 Corsair II                Light Attack     1967
Grumman F-14 Tomcat              Fleet Defense     1974
Sukhoi Su-33                   Carrier Fighter     1998

                     Radius (nm) Source
F-35C Lightning II
Dassault Rafale M
Grumman A-6 Intruder
LTV A-7 Corsair II
Grumman F-14 Tomcat
Sukhoi Su-33
```

```
================================================================================
```

Table 4: Comparative Wing Analysis

```
                             F/A-18E/F (Baseline) F-35C Rafale M A-6 Intruder  \
Reference Area (S) [ft^2]
Aspect Ratio (AR) [-]
Wing Span (b) [ft]
Taper Ratio (λ) [-]
Sweep (Leading Edge) [deg]
Thickness-to-Chord (t/c) [%]
Wing Loading (W/S) [lb/ft^2]

                             A-7 Corsair II F-14 Tomcat Su-33
Reference Area (S) [ft^2]
Aspect Ratio (AR) [-]
Wing Span (b) [ft]
```

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```
Taper Ratio (λ) [-]
Sweep (Leading Edge) [deg]
Thickness-to-Chord (t/c) [%]
Wing Loading (W/S) [lb/ft^2]
```

CHAPTER

TWO

ASSIGNMENT 2 - SYSTEM REQUIREMENTS

Reference: 2025-2026 Undergraduate Team Aircraft Design RFP

Tools Required: MATLAB Requirements Toolbox

2.1 1. Overview

In the last assignment, you researched at the baseline aircraft and worked gathering historical data. Now, you need to define exactly what your *new* aircraft needs to do.

The RFP is just a document full of text from the custumers with their needs. Your goal in this assignment is to a **Requirements Matrix**. This 'matrix' will evolve along the semester and your goal is that your final plane meets every item on this list.

2.2 2. Goals

1. **Find the Explicity Requirements:** Read the RFP and pull out every specific constraint (like "Range > 700 nm" or "Length < 50 ft").
2. **Organize Them:** Group these requirements into categories so they make sense.
3. **List Your Parts:** Define the main physical parts of your airplane (Wing, Engine, Landing Gear, etc.).
4. **Connect the Dots:** Link each rule to the specific part of the aircraft that handles it.

2.3 3. Instructions

2.3.1 Task A: Create the Requirements List (`.slreqx`)

Open the **Requirements Editor** in MATLAB. Create a new set and populate it with the requirements derived from the RFP.

1. **Structure:** Use "Parents" for big categories and "Children" for specific numbers.
   - *Parent:* Carrier Suitability
   - *Child:* Launch Wind Over Deck (0 knots)
2. **Details:** For each requirement, fill in:
   - **Summary:** A short name (e.g., "Max Speed").

- **Description:** The actual text from the RFP.
- **Rationale:** A custom field explaining where this came from (e.g., "RFP Section 3.4.1").

3. **Team Requirements:** If your team wants to add extra constraints (like "Must use a specific engine"), add those too and note them as "Team Derived."

2.3.2 Task B: List the Subsystems (`.slreqx`)

Create a second set in MATLAB. This isn't for requirements; it's for subsystems. List the major systems your aircraft will need.

Typical examples:

- **Aerodynamics:** Wing, Flaps, Tails
- **Propulsion:** Engine, Inlets
- **Structures:** Fuselage, Landing Gear, Fold Mechanism
- **Avionics:** Radar, Cockpit
- **Weights:** Mass Properties

2.3.3 Task C: Connect Requirements to Subsystems (`.slmx`)

Create a **Traceability Matrix**. This is just a Table that connects Task A to Task B.

1. Put Subsystems on top (Columns).
2. Put Requirements on the side (Rows).
3. **Draw Links:** If a subsystem helps meet a requirement, link them.
   - *Example:* "Wingspan < 60 ft" -> Links to **Aerodynamics** (Wing size) and **Structures** (Folding wing).

2.4 4. Deliverables

Submit these three files to Brightspace (replace `XX` with your team number):

1. **TeamXX_Requirements.slreqx**: The checklist of rules.
2. **TeamXX_Subsystems.slreqx**: The list of aircraft parts.
3. **TeamXX_Links.slmx**: The file showing how they connect.

2.5 Appendix: Example List

The code below lists several requirements from the RFP.

Note: *The format and the list are not completed and must be used just as the starting point.*

```
import pandas as pd

# Key rules extracted from RFP Section 3
rfp_data = [
    ("3.1.a", "Configuration", "Any combination of features allowed."),
    ("3.1.b", "Propulsion", "Must use an existing production engine."),
    ("3.1.c", "Materials", "Must survive 25 years in salt/maritime environment."),
    ("3.1.d", "Subsystems", "All tech must be TRL 6 or higher."),
    ("3.2.a", "Carriers", "Must operate from CVN-68 and CVN-78."),
    ("3.2.1.a", "Launch WOD", "Launch on tropical day (89.8°F) with 0 kts wind."),
    ("3.2.1.b", "Catapult Endspeed", "Meet sink rate, lift, and accel limits."),
    ("3.2.2.b", "Approach Speed", "Less than 145 knots."),
    ("3.3.1.a", "Avionics Weight", "2,500 lb internal weight allocation."),
    ("3.3.1.b", "Weapons Load", "6x AIM-120C + 2x AIM-9X."),
    ("3.4.1.a", "Range", "Min 700 nm radius (1,000 nm desired)."),
    ("3.4.1.d", "Turn Rate", "Min 8.0 deg/sec sustained @ 20k ft."),
    ("3.4.1.e", "Dash Speed", "Mach 1.6 @ 30k ft (Mach 2.0 desired)."),
    ("3.5.a", "Load Factor", "7.0g at mid-mission weight."),
    ("3.5.g", "Wingspan (Open)", "Max 60 feet."),
    ("3.5.h", "Wingspan (Folded)", "Max 35 feet."),
    ("3.5.i", "Length", "Max 50 feet."),
    ("3.5.j", "Height", "Max 18.5 feet."),
    ("3.5.l", "Max Weight", "Max 90,000 lb."),
]

df_reqs = pd.DataFrame(rfp_data, columns=["ID", "Summary", "Description"])

# Show the table
display(df_reqs)

# To save this as a file to import, uncomment the line below:
# df_reqs.to_csv("Requirements_Starter_List.csv", index=False)
```

```
         ID            Summary  \
0     3.1.a      Configuration   
1     3.1.b         Propulsion   
2     3.1.c          Materials   
3     3.1.d         Subsystems   
4     3.2.a           Carriers   
5   3.2.1.a         Launch WOD   
6   3.2.1.b  Catapult Endspeed   
7   3.2.2.b     Approach Speed   
8   3.3.1.a    Avionics Weight   
9   3.3.1.b       Weapons Load   
10  3.4.1.a              Range   
11  3.4.1.d          Turn Rate   
12  3.4.1.e         Dash Speed   
13    3.5.a        Load Factor   
14    3.5.g    Wingspan (Open)   
15    3.5.h  Wingspan (Folded)   
16    3.5.i             Length   
```

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```
17    3.5.j             Height
18    3.5.l         Max Weight

                                              Description
0                Any combination of features allowed.
1             Must use an existing production engine.
2   Must survive 25 years in salt/maritime environ...
3                   All tech must be TRL 6 or higher.
4                Must operate from CVN-68 and CVN-78.
5    Launch on tropical day (89.8°F) with 0 kts wind.
6             Meet sink rate, lift, and accel limits.
7                                Less than 145 knots.
8                2,500 lb internal weight allocation.
9                            6x AIM-120C + 2x AIM-9X.
10              Min 700 nm radius (1,000 nm desired).
11                Min 8.0 deg/sec sustained @ 20k ft.
12              Mach 1.6 @ 30k ft (Mach 2.0 desired).
13                      7.0g at mid-mission weight.
14                                       Max 60 feet.
15                                       Max 35 feet.
16                                       Max 50 feet.
17                                     Max 18.5 feet.
18                                     Max 90,000 lb.
```

CHAPTER

THREE

ASSIGNMENT 3 - INITIAL SIZING

Topic: Weight Sizing using Raymer's Refined Method
Subject: Carrier-Based Strike Fighter (F/A-18E Super Hornet Derivative)

3.1 1. Objective

Create a conceptual weight sizing estimation code for a carrier-based interdiction fighter. You will calculate the **Design Takeoff Gross Weight** (W_0) and **Empty Weight** (W_e) required to perform a specific mission profile.

You must use **Raymer's Refined Sizing Equation** (Eq 6.1) for the empty weight fraction, adjusted for naval operations, and the **Fuel Fraction Method** for mission fuel requirements.

3.2 2. Problem Statement

The US Navy requires a new strike fighter capable of launching from a range of carriers to perform "Interdiction" missions (deep strike behind enemy lines). The aircraft must carry a heavy ordnance load while maintaining specific maneuverability parameters.

Your Goal: Determine the W_0 required to fulfill this mission.

3.2.1 Design Requirements

- **Payload ($W_{payload}$):** 6,000 lbs (2x GBU-31 JDAM, 2x AIM-120, 2x AIM-9X, Gun Ammo)
- **Crew (W_{crew}):** 215 lbs (1 Pilot + Gear)
- **Configuration:** Carrier-based, Fixed Wing (or simple fold), Afterburning Turbofans.

3.3 3. Theoretical Background

To size the aircraft, you will solve for W_0 using the fundamental sizing equation:

$$W_0 = \frac{W_{crew} + W_{payload}}{1 - \frac{W_f}{W_0} - \frac{W_e}{W_0}}$$

3.3.1 A. Fuel Fraction Method ($\frac{W_f}{W_0}$)

The mission fuel fraction is calculated by multiplying the weight ratios (W_i/W_{i-1}) of all mission segments.

1. **Cruise Segments (Breguet Range Equation):**

$$\frac{W_i}{W_{i-1}} = \exp\left(\frac{-R \cdot C}{V \cdot (L/D)}\right)$$

   Where R is range, C is SFC, V is velocity, and L/D is the lift-to-drag ratio.

2. **Loiter/Combat Segments (Endurance Equation):**

$$\frac{W_i}{W_{i-1}} = \exp\left(\frac{-E \cdot C}{L/D}\right)$$

   Where E is endurance time (in hours).

3.3.2 B. Empty Weight Fraction ($\frac{W_e}{W_0}$)

You will use Raymer's statistical equation for "Jet Fighters" (Eq 6.1), modified for carrier operations.

$$\frac{W_e}{W_0} = \left(a + b \cdot W_0^{c_1} \cdot AR^{c_2} \cdot \left(\frac{T}{W}\right)^{c_3} \cdot \left(\frac{W}{S}\right)^{c_4} \cdot M_{max}^{c_5} \cdot K_{vs}\right) + \text{Carrier Penalty}$$

3.4 4. The Design Mission (Interdiction Profile)

The aircraft is sized for the following "Design Mission":

1. **Warm-up & Takeoff:** Standard Raymer allowance ($W_1/W_0 = 0.970$).
2. **Climb:** Accelerate to cruise altitude ($W_2/W_1 = 0.985$).
3. **Cruise Out:** 400 NM transit to target area.
4. **Combat:** 3 minutes at Maximum Thrust (Afterburner) at Mach 1.2.
5. **Cruise Back:** 400 NM return to carrier.
6. **Loiter:** 20 minutes at sea level (awaiting recovery).
7. **Landing:** Standard Raymer allowance ($W_7/W_6 = 0.995$).

3.5 5. Technical Data & Assumptions

3.5.1 A. Propulsion: General Electric F414-GE-400

We assume a twin-engine configuration using the F414 turbofan.

Flight Phase	Assumed SFC ($lb/lbf \cdot hr$)	Justification
Cruise	**0.82**	**Nicolai** - Table J.1
Combat	**1.844**	**Max Afterburner.** Fuel consumption more than doubles when reheating exhaust gas for supersonic combat maneuvering.
Loiter	**0.80**	**High Efficiency.** Engine operates at lower RPM/thrust setting, closer to its optimum efficiency point for fuel conservation.

3.5.2 B. Aerodynamics: F/A-18E Platform

The Lift-to-Drag (L/D) ratios are estimated based on the F/A-18E's external store configuration.

Flight Phase	Assumed L/D	Justification
Cruise	**9.2**	**"Dirty" Configuration.** The F-18 carries external pylons, missiles, and sensor pods. While the clean airframe might achieve ~11-12, the parasitic drag from stores significantly reduces L/D during transit.
Combat	**4.5**	**Induced Drag Penalty.** During high-G maneuvering (Turn Rate > 15 deg/sec), the aircraft pulls high Angle of Attack (AOA). Induced drag scales with C_L^2, causing L/D to plummet.
Loiter	**11.0**	**Max Efficiency.** Loiter is flown at $(L/D)_{max}$ speed. We assume a slightly cleaner aerodynamic profile (fuel tanks might be empty/dropped, or speed is optimized) compared to high-speed cruise.

3.5.3 C. Raymer Sizing Constants (Jet Fighter)

- $A = -0.02$
- $B = 2.16$
- $C_1 = -0.10$ (Scale efficiency)
- $C_2 = 0.20$ (Aspect Ratio penalty)
- $C_3 = 0.04$ (Propulsion weight penalty)
- $C_4 = -0.10$ (Wing loading benefit)
- $C_5 = 0.08$ (Mach structural penalty)
- **Carrier Penalty: +0.03 < CP < 0.06** (Must be added to the result to account for keel beam, arresting hook, and heavy gear).

3.5.4 D. Performance Constraints

Use the data collected in the previous Assignment for the F/A-18E Super Hornet

- M_{max} : ________
- **Aspect Ratio** (AR)**:** ________
- **Thrust-to-Weight** (T/W)**:** __________
- **Wing Loading** (W/S)**:** ____________ lb/ft^2

3.6 6. Deliverables

1. **Code:** A Python or Matlab script calculation showing the convergence of W_0.
2. **Slides:** Report the calculated Design Takeoff Gross Weight (W_0) and Empty Weight (W_e). Make sure to describe all assumptions.
3. **Validation:** Use your sizing code to validade your result to the actual F/A-18E Empty Weight (~32,100 lbs). Add it to your slides and discuss the required 'fudge factors' to validate your code.

CHAPTER

FOUR

ASSIGNMENT 4 - AERODYNAMIC ANALYSIS

This assignment explores the aerodynamic modeling of the F/A-18 Hornet, focusing on the high-lift capabilities provided by its Leading Edge Vortex and its efficiency across the subsonic and supersonic flight regimes.

4.1 Part 1: Subsonic Analysis & Vortex Lift

The F/A-18 is famous for its large **Leading Edge Extensions (LEX)**. These generate high-energy vortices that flow over the main wing, providing "vortex lift" and allowing the aircraft to maintain control at very high angles of attack.

- **C_L vs. α Curve:** Create a plot showing the lift coefficient as a function of the angle of attack. You must identify the linear region and the non-linear contribution of the leading edge vortex.
- **Drag Polar Comparison:** Estimate the aerodynamic coefficients and plot the drag polar for the subsonic regime. You must compare the drag polar **with** the leading edge vortex effect versus **without** it to demonstrate the impact on induced drag and stall margins.

4.2 Part 2: Parasite Drag Estimation ($C_{D,0}$)

In this section, you will perform a high-fidelity drag estimate using the **Component Build-up Method**. This requires summing the zero-lift drag of each major part of the airframe.

- **Methodology:** Following the lecture notes, calculate the contribution for the fuselage, wings, tails, and nacelles.
- **Factors:** Your analysis must account for the flat-plate skin friction, form factors (pressure drag), and interference factors (aerodynamic interactions between joined components).

4.3 Part 3: Supersonic Effects and Cruise Efficiency

As the aircraft approaches the speed of sound, compressibility effects significantly change its performance. This part focuses on the "Drag Rise" and how it influences long-range flight.

- **Drag Rise:** Calculate the Drag Divergence Mach number (M_{dd}) and the subsequent wake drag ($C_{D,wake}$).
- **Performance Metrics:** Calculate the induced drag factor as a function of Mach number.
- **Required Graphical Analysis:**
  1. **$C_{D,0} + C_{D,wake}$ vs. Mach:** Illustrate the total zero-lift drag increase as the aircraft moves into the transonic regime.

2. **Maximum L/D vs. Mach:** Show the degradation of maximum lift-to-drag ratio as wave drag appears.
3. $L/D \cdot$ **Mach vs. Mach:** Plot this parameter to identify the aircraft's most efficient cruising speed (Range Factor).

4.4 Deliverables

1. **Code:** A Python or Matlab script calculation required for Parts 1,2 and 3.
2. **Slides:** Summary with all plots and assumptions.
3. **Validation:** Use your code to validade your result to the actual F/A-18E. This code will serve as base for your aerodynamic modeling.

CHAPTER

FIVE

ASSIGNMENT 5 - CONSTRAINT DIAGRAM

Reference Text: Mattingly et al., *Aircraft Engine Design*, Chapter 2.

5.1 1. Overview

The objective of this assignment is to define the "Feasible Design Space" for the Next Generation Carrier-based Strike Fighter. By applying the energy-maneuverability methods from Chapter 2 of Mattingly and analyzing the AIAA Request for Proposal (RFP), you will construct a **Constraint Diagram** (Thrust-to-Weight Ratio T_{SL}/W_{TO} vs. Wing Loading W_{TO}/S).:

If this analysis, your team will determine the initial design point (T/W, W/S) that allows your aircraft to meet all critical performance requirements.

5.2 2. Requirements Extraction

Review Section 3 of the AIAA RFP. You must extract the numeric constraints for the following flight phases and map them to the corresponding "Case" equations in Mattingly Chapter 2.

Constraint	RFP Section	Mattingly Case	Critical Value(s) to Find
Carrier Landing	3.2.2	Case 11 (Eq 2.48)	Max Approach Speed (V_{app}), Stall Margin
Carrier Takeoff	3.2.1	Case 10 (Eq 2.45)	End Speed (V_{end}), Wind Over Deck (WOD)
Supersonic Dash	3.4.1.e	Case 1 (Eq 2.12)	Mach 1.6 @ 30k ft ($P_s = 0$)
Strike Dash	3.4.2.c	Case 1 (Eq 2.12)	Mach 0.85 @ SL (**Dry Thrust**)
Sustained Turn	3.4.1.d	Case 3 (Eq 2.15)	8.0 deg/sec @ 20k ft ($P_s = 0$)
SEROC (Climb)	3.5.f	Case 2 (Eq 2.14)	500 ft/min @ Approach (OEI)

The table above shows some examples of the link between the performance requirements and the equations used to generate the constraint diagram.

5.3 3. Engineering Assumptions

In this preliminary phase, you do not yet have a drag polar or an engine deck. You must use the functions you created for your Assignment 4 to create the drag polar and use the thrust lapse ration using Mattingly Eq 2.54a. **You must justify all selected values in your final report.**

5.3.1 A. Aerodynamics (The "Loaded" Polar)

The RFP requires ordnance to be carried for the entire mission. You cannot use "clean" drag values.

- **Subsonic (**$M < 1.0$**):** Estimate C_{D0} and K_1 for a loaded configuration (e.g., missiles/bombs external). Estimate $C_{L_{max}}$ for Takeoff (flaps) and Landing (full high-lift).
- **Supersonic (**$M \geq 1.0$**):** Estimate C_{D0} and K_1 for supersonic flight. **Crucial:** You must increase C_{D0} to account for wave drag and increase K_1 to account for efficiency loss.

5.3.2 B. Weight Fractions ($\beta = W/W_{TO}$)

Constraints are evaluated at instantaneous weights. Estimate β for each phase:

- **Takeoff:** $\beta = 1.0$
- **Combat/Turn:** $\beta \approx 0.78$ (Mid-mission)
- **Landing:** $\beta \approx 0.60$ (See RFP for fuel reserve/payload bring-back specifics)

5.3.3 C. Propulsion ($\alpha = T/T_{SL}$)

Model the installed thrust lapse (α) as a function of Altitude and Mach. Use Mattingly Eq 2.54a (Low Bypass Turbofan) or similar.

- **Wet Thrust (**α_{wet}**):** Use for Supersonic Dash and Sustained Turn.
- **Dry Thrust (**α_{dry}**):** Use for the Strike Dash ($M = 0.85$ @ SL).
- **OEI Thrust:** For the SEROC constraint, available thrust is 50% of total (Twin Engine).

5.4 4. Analysis Tasks

5.4.1 Task A: Carrier Limits

1. **Landing Constraint:** Calculate max W_{TO}/S for 145 kt approach. (Don't forget to account for $\beta_{landing}$!)
2. **Takeoff Constraint:** Calculate max W_{TO}/S for a C-13-2 catapult launch ($V_{end} \approx 165$ kts, $WOD = 0$).

5.4.2 Task B: Performance Limits

Implement the "Master Equation" (Mattingly Eq 2.11) to plot the required T_{SL}/W_{TO} across a range of wing loadings (e.g., 40 to 140 lb/ft^2).

1. **Supersonic Dash:** Mach 1.6 @ 30k ft (Wet Thrust, Supersonic Drag).
2. **Strike Dash:** Mach 0.85 @ Sea Level (**Dry Thrust**, Loaded Subsonic Drag).
3. **Sustained Turn:** 8 deg/sec @ 20k ft (Wet Thrust). *Optimization:* Calculate required thrust at M 0.7, 0.8, and 0.9 to find the minimum.
4. **SEROC:** 500 ft/min climb rate on **One Engine** in approach configuration (Gear Down/Flaps Down -> High Drag).
5. …

5.5 Deliverables

1. **Code:** A Python or Matlab script containing the calculations required. Ensure your scripts are capable of iterating through various wing and thrust loading parameters.
2. **Slides:** A summary presentation featuring all generated plots. It is required one slide with the **List of Assumptions**. You must explicitly list and justify the values used for:
   - **Aerodynamic Coefficients:** Zero-lift drag ($C_{D,0}$ and $C_{D,w}$) and Oswald efficiency factor (e).
   - **Weight Fractions:** Estimated fuel and payload fractions for the specific mission profile.
   - **Environmental Constants:** Atmospheric density and altitude-specific performance adjustments.
   - **Propulsion:** Assumed thrust lapse ratio.
3. **Design Point Selection:** Based on the **feasible design space** ,identified in your constraint diagram, select and justify a specific design point (W/S and T/W).
   - **Feasibility Analysis:** Clearly highlight the region of the plot where all mission constraints (e.g., takeoff distance, cruise speed, service ceiling) are satisfied.
   - **Optimization Logic:** Explain why you chose your specific point within that region. Are you optimizing for the smallest wing area (lowest weight) or higher thrust for better maneuverability?
   - **Constraint Sensitivity:** Identify which constraint is the "active" or "critical" one that most limits your design space.

CHAPTER

SIX

ASSIGNMENT 6 - DESIGN CHECK 1 – PROPULSION INTEGRATION & WING DEFINITION

6.1 1. Objective

The goal of this assignment is to freeze the primary configuration parameters of your aircraft. Teams will transition from "unknown engine" sizing to a realistic design by selecting an off-the-shelf propulsion system and defining the specific wing geometry. You will use your Constraint Diagram to verify that the selected engine and wing area meet all mission requirements and update your Maximum Takeoff Weight (MTOW) based on these physical selections.

6.2 2. Tasks

6.2.1 Task 2.1: Propulsion System Selection

Using the **Constraint Diagram** developed in the previous assignment:

1. **Identify the Design Space:** Locate your feasible design space (the area satisfying all T/W and W/S constraints).
2. **Select an Engine:** Search for commercially available "off-the-shelf" propulsion systems (e.g., Rotax, Lycoming, Pratt & Whitney, GE, JetCat, etc.) that fit your aircraft class.
3. **Determine Number of Engines:** Decide on the number of engines (N_{eng}) required to meet the Thrust-to-Weight (T/W) requirement found in your constraint diagram.
4. **Verification:** Plot the installed thrust of your selected propulsion system (considering approximate installation losses, e.g., 3-5% for inlets/nozzles) onto your Constraint Diagram to ensure the point lies within the feasible design space.

6.2.2 Task 2.2: Wing Planform & Airfoil Selection

Define the physical geometry of the wing:

1. **Wing Loading (W/S):** Select a specific design point (W/S) from your constraint diagram that minimizes weight or cost while meeting all constraints.
2. **Wing Area (S):** Calculate the required wing area based on your current W_{TO} estimate and selected W/S.
3. **Planform Parameters:** Select and justify the following:
   - Aspect Ratio (AR)
   - Sweep Angle (Λ)

- Taper Ratio (λ)

4. **Airfoil Selection:** Select an airfoil that meets your $C_{L_{max}}$ requirements (landing/takeoff) and cruise drag characteristics. Include a lift-curve slope and drag polar estimation for the selected airfoil and for the defined wing planform using OpenVSP, XFLR5 ,or a similar software. Make sure to add the effect of the leading-edge vortex.

6.2.3 Task 2.3: Weight Refinement (Sensitivity Update)

The selection of a specific engine and wing geometry changes the aircraft weight. You must now determine the critical sizing case.

1. **Engine Weight:** Replace the statistical engine weight estimation with the *actual* dry weight of your selected engine(s) + accessories.
2. **Wing Weight:** Estimate the wing weight impact based on your selected Aspect Ratio and Sweep angle (higher AR and Sweep generally increase structural weight).
3. **Dual-Mission Sizing:** Perform the weight sizing analysis for **both** missions defined in the RFP to determine which one drives the design (the "Design Mission"):
   - **Air-to-Air Combat Mission:** (RFP Sections 3.3.1 & 3.4.1) – Note the dash requirement of Mach 1.6 at 30,000 ft and sustained turn requirements.
   - **Strike Mission:** (RFP Sections 3.3.2 & 3.4.2) – Note the sea level dash requirement (Mach 0.85+) and specific payload configuration.
4. **New MTOW:** Recalculate the Maximum Takeoff Weight (W_{TO}) based on the most demanding of the two missions above.
   - *Check:* If the new weight varies significantly (>5%) from your previous iteration, you must re-check if your selected engine still provides enough Thrust/Weight.

6.3 3. Deliverables

Submit a presentation and the input file, including the following sections:

6.3.1 3.1 Input Parameters Table

Provide a clear table listing the inputs used for your decisions:

- **Mission Inputs:** Required Range/Endurance, Payload Weight.
- **Constraint Inputs:** $C_{L_{max}}$ used, field lengths required.
- **High Lift Devices:** Selection of flaps/slats types (e.g., plain, split, fowler) and anticipated deflection angles for takeoff and landing.
- **Selected Geometric Inputs:** Aspect Ratio (AR), Taper (λ), Sweep (Λ), Thickness-to-chord ratio (t/c).

6.3.2 3.2 Propulsion & Wing Output Report

- **Propulsion Selection:**
  - Manufacturer and Model Name.
  - Max Thrust Dry/MIL (Sea Level Static).
  - Number of Engines.
  - SFC Dry/MIL.
  - Engine Weight.
- **Wing Definition:**
  - Selected Airfoil Section (include an image of the profile).
  - Final Wing Area (S_{ref}).
  - Wingspan (b).
  - Mean Aerodynamic Chord (MAC).
- **Performance Verification:**
  - **Thrust Available vs. Thrust Required:** A value comparison showing your selected engine exceeds the requirement.
  - **Final Design Point:** An updated Constraint Diagram showing the specific point selected (T/W and W/S).

6.3.3 3.3 Weight Update

- **New Maximum Takeoff Weight** (W_{TO})
- **Empty Weight** (W_{empty})
- **Fuel Weight** (W_{fuel})

6.3.4 3.4 Configuration "Single Source of Truth" File

Submit a digital configuration file (choose either a **MATLAB .m** script or a **JSON** file) that serves as the central database for your aircraft parameters.

- **Content:** This file must list every input variable, constant, and assumed value used in your design code (e.g., C_{D0}, AR, $W_{payload}$, SFC …).
- **Justification:** Every parameter must be documented with a comment (MATLAB) or a description field (JSON) explaining the **source or justification** for that value (e.g., *"Derived from ray theory estimation"* or *"From Rotax 912 User Manual"*).
- **Version Control:**
  - Include a header or metadata field tracking the **Version Number** (e.g., `v1.2`).
  - **Change Log:** If a value has changed from a previous assignment (e.g., updating statistical engine weight to actual engine weight), you must explicitly state the **reason for the change** in the file comments.

CHAPTER

SEVEN

ASSIGNMENT 7 - CONFIGURATION GENERATION AND SELECTION

7.1 1. Purpose

The primary objective of this phase is to transition from a wide array of divergent concepts to few configurations that satisfies the requirements of the 2025-2026 AIAA RFP.

7.2 2. Part A: Morphological Chart

Create a chart to explore your design space based on required fucntionalities and components definitions.

- **List Functions**: Identify 6–8 primary functions (e.g., Lift, Stability, Weapons Carriage, Propulsion).
- **Component Definitions**: Fill each row with different technical ways to fulfill that function (e.g., Single vs. Twin Engine, V-tail vs. Conventional tail).
- **Combine**: Select one option from each row to form **distinct aircraft concepts**.

7.3 3. Part B: Brainstorming

Meet as a team to refine your initial concepts and generate new ideas.

- **Rules**: No criticism of ideas is allowed; focus on generating a large quantity of "wild" ideas to ensure no promising concept is missed.
- **Constraint Focus**: For instance, brainstorm how to fit a 60ft wingspan into a 35ft folded space and keep the total length under 50ft to meet shipboard limits.

7.4 4. Part C: The 10-Step Pugh Method

Evaluate your concepts against a baseline.

Step	Task
1. Choose Criteria	Select critcal metrics from the requirements (e.g., 700nm radius, carrier approach speed).
2. Form Matrix	Set up your evaluation grid with criteria on the left and concepts across the top.
3. Clarify Concepts	Ensure every team member understands how each design works to develop team ownership.
4. Choose Datum	Pick the "Datum" (reference aircraft) to compare others against.
5. Run Matrix	Rate concepts as **+** (better), **-** (worse), or **S** (same) compared to the Datum.
6. Evaluate	Total the pluses and minuses separately; do not calculate a single final score.
7. Attack Negatives	Modify designs to fix "-" ratings and borrow "+" features for other designs.
8. Rerun Matrix	Select a new promising design as the Datum and repeat the evaluation.
9. Plan Work	Decide what math or research is needed to resolve uncertain ratings.
10. Iterate	Repeat the process until your team converges on one winning design.

7.5 5. Part D: CAD Modeling (OML)

Create Outer Mold Line (OML) models for your **top three concepts** to verify physical feasibility.

- **Size Check**: Ensure the 2,500 lb avionics suite and internal weapons fit inside the proposed airframe.
- **Fuel Volume Check**: Determine the maximum fuel volume available considering the wing and fuselage.
- **Carrier Fit**: Confirm the plane is not taller than 18.5ft and meets the 35ft folded wingspan limit for hangar storage.

7.6 6. Deliverables (Presentation Format)

Prepare your findings as a slide deck including:

1. **Morphological Chart –** Your table of functions and technical solution options.
2. **Candidate Concepts –** Sketches (make it professional) and descriptions of your initial designs.
3. **Selection Matrix –** Your final Pugh Matrix showing scores and iteration history.
4. **OML Verification –** CAD screenshots proving the top three designs fit carrier height and length limits.
5. **The Selected Configuration –** A summary of why the chosen(s) design(s) are the best fit to respond to the RFP.
6. **Questions to answer** - For the selected configuration(s):

- Does our chosen wing shape allow for a 60ft wingspan that can reliably fold down to the required 35ft limit for shipboard storage?
- Does our tail configuration provide necessary stability while ensuring the aircraft remains under the 18.5ft maximum height limit?
- Does the inlet geometry maximize airflow efficiency without compromising the internal volume needed for fuel and the wing structure?
- Does our choice of engine count (Single vs. Twin) leave sufficient internal space to house the required 2,500 lb avionics suite and internal weapons?
- Does this configuration balance the need for a 700nm radius with the carrier approach speed requirements?

CHAPTER

EIGHT

ASSIGNMENT 8 - HORIZONTAL TAIL SIZING AND TRIM ANALYSIS

8.1 1. Purpose

The objective of this assignment is to size the horizontal tail of your aircraft to satisfy both static longitudinal stability and control requirements (such as rotation at takeoff or stall recovery). Additionally, you will analyze the longitudinal trim conditions to solve for control surface deflections and quantify the "Trim Drag" penalty associated with maintaining level flight. Finally, you will establish the initial sizing for the vertical tail based on directional stability criteria.

8.2 2. Task Description

8.2.1 Part A: Horizontal Tail Geometry & Volume Coefficients

1. **Geometric Layout:** Define your horizontal tail planform area (S_t), span (b_t), and mean aerodynamic chord ($\bar{c}_t$). Specify the airfoil choice, aspect ratio (AR_t), and sweep (Λ_t).
2. **Tail Arm:** Determine the tail arm (l_t), defined as the distance between the wing and tail aerodynamic centers.
3. **Volume Coefficient:** Calculate your horizontal tail volume coefficient (V_H).

$$V_H = \frac{S_t l_t}{S_w \bar{c}}$$

4. **Benchmarking:** Compare your calculated V_H to historical aircraft data.

8.2.2 Part B: Scissor Plot

1. **Define Boundaries:** Calculate the following limits to construct your scissor plot:
   - **Stability (Aft Limit):** Determine the tail area required to maintain your defined static margin ($SM > 0$).
   - **Stall Recovery (Aft Limit):** Calculate the area needed to provide sufficient nose-down pitching moment for recovery at $C_{L_{max}}$.
   - **Take-off Rotation (Forward Limit):** Calculate the area required to provide the nose-up moment necessary to rotate the aircraft at the main landing gear position.
2. **Sizing Validation:** Identify the "Intersection Point" on your plot where all stability and control requirements are met. Overlay your selected S_h/S to define your feasible CG range.

8.2.3 Part C: Trim Analysis & Trimmed Drag Polar

1. **Trim Equations:** Solve the system of two equations for the two unknowns: trim angle of attack (α_{trim}) and trim elevator deflection ($\delta_{e_{trim}}$).

$$\begin{cases} C_L = C_{L_0} + C_{L_\alpha}\alpha_{trim} + C_{L_{\delta_e}}\delta_{e_{trim}} \\ C_M = 0 = C_{M_0} + C_{M_\alpha}\alpha_{trim} + C_{M_{\delta_e}}\delta_{e_{trim}} \end{cases}$$

2. **Trimmed Drag Polar:** Generate a plot of the Trimmed Drag Polar (C_D vs $C_{L_{trim}}$) and compare it against the "Clean" (untrimmed) polar.
3. **Quantify Penalty:** Account for the drag resulting from elevator deflection and the additional wing lift required to offset the "tail download" (negative lift on the tail).

$$C_D = C_{D_0} + KC_L^2 + \frac{S_t}{S}\frac{C_{L_t}^2}{\pi e_t AR_t}$$

8.2.4 Part D: Vertical Tail Geometry & Volume Coefficient

1. **Geometric Layout:** Define your vertical tail planform area (S_v), span/height (b_v), and aerodynamic chord parameters (c_v^{root}, c_v^{tip}). Specify the airfoil choice, aspect ratio (AR_v), and sweep (Λ_v).
2. **Vertical Tail Arm:** Determine the vertical tail arm (l_v), the longitudinal distance from the wing quarter chord to the vertical tail aerodynamic center.
3. **Directional Volume Coefficient:** Calculate the vertical tail volume coefficient (V_v) using the wing span (b_w) as the reference length.

$$V_v = \frac{S_v l_v}{S_w b_w}$$

4. **Benchmarking:** Compare your V_v to typical values for your aircraft category to ensure sufficient directional stability and rudder authority.

8.3 3. Deliverables

Submit a presentation including:

- **Tail Design Table:** Summary of all geometric parameters ($S_t, S_v, l_t, l_v \ldots$) and volume coefficients (V_H, V_v).
- **The Scissor Plot:** A plot clearly labeled with stability and control limits.
- **Trim Plots:** Graphs showing α_{trim} and $\delta_{e_{trim}}$ vs. $C_{L_{trim}}$.
- **The Trimmed Polar:** A comparison of C_D vs. C_L for trimmed and clean configurations.
- **Discussion:** Using XFLR5, OpenVSP or AVL analyze if your current tail sizing (horizontal and vertical) provides a sufficient CG range and directional stability for your aircraft's mission profile.

CHAPTER

NINE

ASSIGNMENT 9 - PERFORMANCE ASSESSMENT

9.1 Objective

The objective of this assignemnt is to simulate and evaluate the performance of your candidate strike fighter aircraft and verify compliance with the Request for Proposal (RFP) requirements.

9.2 Description

Develop a Matlab or a Python code to analyze the performance characteristics of your aircraft design. Use the mathematical models discussed in class to define the drag polar, thrust lapse, and point mass equations of motion.

9.3 Tasks

9.3.1 1. Carrier Takeoff Performance

Simulate the aircraft takeoff roll on a carrier deck to verify launch capability.

- Model the aircraft dynamics using Newton's Second Law to calculate acceleration along the deck.
- Include the energy provided by the C-13-2 catapult (assume 60.5 million ft-lbf over a 310 ft stroke) and calculate the average catapult force.
- Evaluate the takeoff on a tropical day with zero Wind Over Deck (WOD).
- **Plot:** Airspeed versus distance along the deck.
- **Output:** Determine the final airspeed at the end of the catapult stroke and verify if it exceeds the required takeoff safety speed ($1.2V_{stall}$).

9.3.2 2. Maneuvering Performance (Doghouse Plot)

Evaluate the turn performance at an altitude of 20,000 ft.

- Calculate the aerodynamic limit (wing stall), structural limit (maximum load factor), and thrust limit ($P_s = 0$).
- Calculate the Instantaneous Turn Rate (ITR) and Sustained Turn Rate (STR).
- Calculate the Corner Speed (V^*).
- **Plot:** Doghouse plot showing Turn Rate versus Mach number. Include contours for constant turn radius and constant load factor.

- **Output:** Verify if the aircraft meets the 8.0 deg/sec minimum sustained turn rate requirement at mid-mission weight.

9.3.3 3. Flight Envelope and Specific Excess Power

Map the sustained operating envelope of your aircraft.

- Calculate the Specific Excess Power (P_s) over a grid of Mach numbers (0.1 to 2.5) and altitudes (0 to 65,000 ft).
- Implement a Mach-dependent drag polar to account for the transonic wave drag rise.
- Implement the thermodynamic thrust lapse model to estimate available thrust at altitude.
- **Plot:** $P_s = 0$ contours for load factors from $1g$ up to the maximum structural limit (e.g., $7.5g$).

9.3.4 4. Impact of External Stores

Analyze how the strike mission payload affects the flight envelope.

- Add the estimated weight and parasite drag (ΔC_{D0}) for the external ordnance specified in the RFP (4x JDAMs, 2x AIM-9X, and pylons), also check the impact of the external fuel tanks.
- **Plot:** Overlay the $1g$ $P_s = 0$ envelope of the loaded strike configuration directly on top of the clean configuration envelope for comparison.

9.3.5 5. Carrier Landing and Arrestment

Verify the aircraft can safely recover on the carrier.

- Calculate the approach speed and the engaging speed relative to the ship, assuming a 15-knot WOD.
- **Plot:** The kinetic energy limit curve of the Mark 7 Mod 3 arresting gear (Airplane Hookload vs. Engaging Speed) and plot your aircraft's specific touchdown condition.
- **Output:** Confirm if the landing weight and engaging speed fall within the safe operating envelope of the arresting gear.

9.3.6 6. Single Engine Rate of Climb (SEROC)

Evaluate the safety margins in the event of an engine failure.

- Calculate the rate of climb with one engine inoperative for the launch configuration (gear down) and the approach configuration (gear and full flaps down).
- **Output:** Verify if the aircraft meets the minimum SEROC requirements for launch and for approach.

9.3.7 7. Climb Performance: Rate and Angle of Climb

Evaluate the climbing capabilities of your aircraft at 10,000 ft.

- Calculate the Rate of Climb (ROC) and Angle of Climb (AOC) for the clean configuration.
- Identify the Best Angle of Climb Speed (V_x) and Best Rate of Climb Speed (V_y).
- **Plot:** A dual-axis plot showing Rate of Climb and Angle of Climb versus True Airspeed, explicitly marking V_x and V_y.
- **Output:** Discuss the operational implications of your calculated V_x and V_y speeds.

9.3.8 8. Payload-Range Envelope

Determine the strategic reach of your aircraft by constructing a payload-range diagram.

- Use the Breguet Range Equation to calculate the range for different payload and fuel combinations.
- Evaluate two configurations: a clean aircraft relying only on internal fuel, and a loaded aircraft utilizing external drop tanks (accounting for the added empty weight and increased parasite drag).
- **Plot:** Payload-Range Envelope comparing both the internal fuel only and external tank configurations.
- **Output:** Verify if the aircraft meets the minimum 700 nm combat radius requirement specified in the RFP.

9.4 Deliverables

Submit a presentation slide deck (e.g., PowerPoint or PDF) via the course portal. The presentation must contain:

1. **Performance Plots:** The generated plots for each of the 8 tasks, ensuring proper labels, legends, and units are displayed clearly.
2. **Analysis & RFP Verification:** A slide (or section) for each task discussing the results, interpreting the graphs, and explicitly concluding whether your aircraft met the associated RFP requirements.
3. **Overall Assessment:** A final conclusion slide summarizing the viability of the design and highlighting any critical performance shortfalls or necessary design trades.

CHAPTER

TEN

ASSIGNMENT 10 - STRUCTURAL LOADS AND V-N DIAGRAM

10.1 Objective

The objective of this assignment is to define the structural operating envelope (V-n diagram) accounting for different operating altitudes, estimate the inertial and aerodynamic loads on the fuselage for a balanced aircraft trim case, and determine the wing box load distribution.

10.2 Description

Develop a mathematical model (using MATLAB or Python) to compute the limit load factors and generate the V-n diagram using a International Standard Atmosphere (ISA) model. In addition, develop a solver for the fuselage loads that balances the full aircraft weight against aerodynamic lift. Finally, calculate the aerodynamic loads over the wing.

10.3 Tasks

10.3.1 1. Dynamic V-n Diagram (Mach vs. Load Factor)

Construct the V-n diagram for your aircraft .

- Implement an International Standard Atmosphere (ISA) model to dynamically calculate static pressure (p) as a function of geometric altitude (h).
- Map dynamic pressure (q) directly from Mach number (M) to establish the load factor limits (n_{stall}).
- Close the low-speed side of the envelope using the stall speed for the load factor 1.0.
- Consider the positive and negative structural limits (n_{max} and n_{min}).
- **Plot:** Plot the maneuver V-n diagram, considering altitudes of 5000m, 10000m, and 15000m with appropriate envelope curves.

10.3.2 2. Fuselage Loads (Balanced Assembly Case)

Estimate the longitudinal load distribution along the fuselage for a critical pull-up maneuver point on the V-n diagram (e.g., $n_z = 7.5$).

- Distribute the fuselage shell structural mass from nose to tail.
- Incorporate concentrated system/point masses (e.g., Radar, Nose Gear, Avionics, Wing, Engines, etc.) mapped to specific Fuselage Stations (FS).
- Implement a Global Trim condition to find the balance of aerodynamic lift at the wing (L_w) and tail (L_t) required to counteract the total inertial weight at n_z.
- Integrate the distributed structural weight, point masses, and balancing aerodynamic forces from the nose aftwards to calculate shear force (V) and bending moment (M).
- **Plot:** Generate professional, dual-axis shear force and bending moment diagrams along the fuselage length, indicating key stations such as the CG and aerodynamic centers.
- **Output:** Identify the location and magnitude of the peak bending moment and shear force on the fuselage.

10.3.3 3. Wing Box Load Distribution

Establish the external aerodynamic loads on the multi-spar wing box during a critical maneuver case.

- Use Schrenk's approximation by integrating an elliptical lift distribution with a planform-proportional lift distribution or the lift obtained from your aerodynamic solver.
- Calculate the internal Shear Force (V) and Bending Moment (M).
- **Plot:** Generate a plot showing the lift distribution (lbf/in), Shear Force (lbf), and Bending Moment ($lbf - in$) along the spanwise coordinate (y).

10.4 Deliverables

Submit a presentation slide deck (e.g., PowerPoint or PDF) via Brightspace. The presentation must contain:

1. **Dynamic V-n Diagram:** The finalized V-n plot mapping Mach to load factor, with limit lines.
2. **Fuselage Load Diagrams:** Shear and bending moment plots indicating maximum values and assembly locations for the balanced case.
3. **Wing Box Load Diagrams:** Lift distribution, shear force, and bending moment mapped along the true spanwise coordinate.
4. **Summary & Discussion:** Discuss how the peak loads impact your structural sizing, and the impact of operating in different altitudes on the flight envelope.

CHAPTER

ELEVEN

ASSIGNMENT 11 - MULTI-SPAR WING BOX STRUCTURAL DESIGN

11.1 Objective

The objective of this assignment is to perform the structural design and analysis of a multi-spar wing box for the aircraft sized in previous assignments. Using the aerodynamic loads calculated in **Assignment 10** (Schrenk's lift distribution, shear force V, bending moment M, and applied torsion T), design the structural members (skins, spar webs, spar caps) and evaluate all critical failure modes. Demonstrate through Margin of Safety (MS) calculations that the wing box meets the structural requirements for a $n_z = 7.5\,g$ ultimate maneuver case.

11.2 Background

A typical multi-spar wing box consists of:

- **Upper and lower skins** that carry bending loads (compression/tension) and contribute to torsion resistance.
- **Spar webs** that carry the vertical shear loads and contribute to torsion resistance.
- **Spar caps** (flanges) that are lumped-area booms carrying axial bending loads at maximum distance from the neutral axis.
- **Ribs** spaced at intervals L_{rib} that maintain the airfoil shape, transfer loads, and set the buckling panel length.

The structural analysis must address the following failure modes:

1. **Bending Yield —** Direct stress from the bending moment must not exceed the material tensile/compressive yield strength.
2. **Shear Flow —** Multi-cell torsional shear flow must be solved using twist-compatibility (Bredt–Batho) equations.
3. **Web Shear Buckling —** The spar web panels (bounded by spar caps and ribs) must not buckle under the combined transverse shear and torsional shear.
4. **Skin Panel Buckling —** The upper skin panels (in compression during positive-g maneuvers) must not buckle.

Material: Aluminum 7075-T6 (isotropic)

Property	Value
Elastic Modulus E	10.4×10^6 psi
Shear Modulus G	3.9×10^6 psi
Poisson's Ratio ν	0.33
Tensile Yield F_{ty}	73,000 psi
Shear Yield F_{sy}	42,000 psi

11.3 Tasks

11.3.1 1. Cross-Section Geometry and Moment of Inertia

Define the wing box cross-section at the root and evaluate it along the span.

- Use a NACA 4-digit thickness distribution (or similar) to obtain the true airfoil heights at each spar location.
- Choose the number of spars (minimum 3, recommended 4), front spar location (e.g. 15% chord), and rear spar location (e.g. 65% chord).
- Choose structural sizing variables: skin thickness t_{skin}, spar web thickness t_{web}, spar cap area A_{cap}, and rib spacing L_{rib}.
- Compute the area moment of inertia I_{xx} of the wing box cross-section using the parallel-axis theorem for skins, webs ($\frac{1}{12}t_{web}h^3$), and caps ($A_{cap}(h/2)^2$).
- **Plot:** I_{xx} variation along the span.

11.3.2 2. Bending Stress and Margin of Safety

Evaluate the maximum bending stress at each spanwise station.

- Using $\sigma_{max} = \frac{M \cdot z_{max}}{I_{xx}}$, compute the peak fiber stress, where z_{max} is the distance from the neutral axis to the outermost fiber.
- Compute the **Margin of Safety** for bending yield: $\mathrm{MS}_{bending} = \frac{F_{ty}}{\sigma_{max}} - 1$.
- All MS values must be ≥ 0 for a successful design.
- **Plot:** Bending stress distribution vs. spanwise station.
- **Output:** Root bending stress and MS.

11.3.3 3. Multi-Cell Torsional Shear Flow

Solve for the constant shear flow in each cell under the applied torsion.

- Compute the enclosed area A_i and skin arc length s_i for each cell of the multi-spar box.
- Set up the twist-compatibility equations (Bredt–Batho): $\frac{1}{2A_iG} \oint \frac{q}{t} ds = \frac{d\theta}{dz} \quad \text{(same twist rate for all cells)}$
- Add the torque-equivalence equation: $T = \sum_i 2A_i q_i$.
- Solve the resulting linear system for the shear flows q_i and twist rate $d\theta/dz$.
- **Plot:** Annotated cross-section of the airfoil showing shear flow directions and magnitudes in each cell.

11.3.4 4. Spar Web Shear Buckling

Evaluate the shear buckling of the most critical spar web panel.

- The web panel dimensions are set by the spar height h and the rib spacing L_{rib}.
- Compute the buckling coefficient k_s for a clamped panel: $k_s = 8.98 + \frac{5.6}{(a/b)^2}$ where $a/b = \max(L_{rib}, h)/\min(L_{rib}, h)$.
- Compute the critical buckling stress: $\tau_{cr} = k_s \frac{\pi^2 E}{12(1-\nu^2)} \left(\frac{t_{web}}{b}\right)^2$ where $b = \min(L_{rib}, h)$.
- The applied shear stress combines the vertical shear (V) component and the torsional shear flow: $\tau_{applied} = (q_{max,torsion} + q_V)/t_{web}$.
- Compute the **Margin of Safety** for web shear buckling: $\text{MS}_{buckling} = \frac{\tau_{cr}}{\tau_{applied}} - 1$.
- Compute the **Margin of Safety** for web material shear yield: $\text{MS}_{yield} = \frac{F_{sy}}{\tau_{applied}} - 1$.
- **Output:** τ_{cr}, $\tau_{applied}$, and both MS values.

11.3.5 5. Spanwise Structural Response

Evaluate the structural response continuously along the semi-span.

- Compute $I_{xx}(y)$, $\sigma_{max}(y)$, deflection $\delta(y)$ (by double-integrating the curvature $M/(EI)$), and twist angle $\theta(y)$ (by integrating the twist rate from the shear flow solution).
- **Plot:** A 2×2 panel plot showing: (a) I_{xx} vs. y, (b) σ_{max} vs. y with yield limit line, (c) deflection vs. y, (d) twist angle vs. y.

11.3.6 6. (Bonus) 3D Wing Box Visualization

Generate a 3D plot of the swept, tapered wing box.

- Show the wing skins (with transparency), internal spar webs, spar caps, and ribs.
- The geometry must correctly account for sweep angle, taper ratio, and true airfoil contour.

11.4 Deliverables

Submit a presentation slide deck (e.g., PowerPoint or PDF) via Brightspace. The presentation must contain:

1. **Cross-section sketch** with labeled structural members and dimensions.
2. **Bending Analysis:** I_{xx}, root bending stress, and MS for bending yield.
3. **Shear Flow Diagram:** Annotated airfoil cross-section with cell shear flows.
4. **Buckling Analysis:** τ_{cr}, $\tau_{applied}$, MS for web buckling and material yield.
5. **Spanwise Response Plots:** I_{xx}, stress, deflection, twist vs. span.
6. **MS Summary Table:** A table listing each failure mode, allowable, applied, and Margin of Safety.
7. **Summary & Discussion:** Discuss whether the design passes all structural checks and, if not, propose design changes to achieve positive margins.

CHAPTER

TWELVE

ASSIGNMENT 12 - DYNAMIC STABILITY

12.1 Objective

The objective of this assignment is to evaluate the dynamic stability characteristics of the aircraft sized in previous assignments. Based on the 2025-2026 UG AIAA RFP, you must demonstrate aircraft stability for all key flight and loading conditions. You will compute the necessary aerodynamic derivatives and analyze the longitudinal and lateral natural modes of your aircraft.

12.2 Background

As discussed in lecture, an aircraft in flight exhibits key natural dynamic modes, divided into longitudinal and lateral dynamics: To analyze these modes (stick-free or stick-fixed), you first need to determine the stability and control derivatives of your specific aircraft configuration.

12.3 Tasks

- Select your preferred aerodynamic solver (e.g., AVL, XFLR5, OpenVSP/VSPAERO, or SUAVE) to compute the stability derivatives.
- Build or import your aircraft geometry into the solver.
- Compute the non-dimensional longitudinal derivatives and lateral-directional derivatives at flight conditions for both configurations (Air to air and Strike).

12.3.1 2. Longitudinal Dynamic Stability

- Solve for the eigenvalues of the longitudinal system.
- Identify the Phugoid and Short-Period modes.
- Calculate the natural frequency (ω_n) and damping ratio (ζ) for both modes.

12.3.2 3. Lateral Dynamic Stability

- Solve for the eigenvalues of the lateral system.
- Identify the Spiral, Roll Damping, and Dutch Roll modes.
- Calculate the time-to-half/double amplitude for the Spiral and Roll modes, and (ω_n, ζ) for the Dutch Roll mode.

12.3.3 4. Stability Evaluation

- Evaluate whether the aircraft possesses adequate dynamic stability.
- If an instability exists, identify whether it falls within acceptable flying qualities guidelines or the design of the active control.

12.4 Deliverables

Submit a presentation slide deck (e.g., PowerPoint or PDF) via Brightspace. The presentation must contain:

1. **Solver Summary:** Images of your aircraft model in your preferred aerodynamic solver and a table of the key calculated stability derivatives.
2. **Longitudinal Analysis:** Eigenvalues, ω_n, and ζ for the Phugoid and Short-Period modes. Include a brief discussion on their stability.
3. **Lateral Analysis:** Eigenvalues, time constants, and Dutch roll parameters. Include a brief discussion on their stability.
4. **Conclusions:** A final assessment of the dynamic stability of your aircraft as required by the RFP, noting any design tweaks (like tail resizing or dihedral changes) needed to fix unstable modes, or the design of the active control.